ILLUMINATED INTEGRATION

FIELD TERMINATION GUIDE

GUIDE
01 / 15

Pinouts, polarity and color codes for AV and lighting installs.

HOUSE RULES

Black is never hot.

Feed-Through RJ45 connectors are not allowed

Audio XLR: **pin 2 = +** · DMX: **pin 3 = Data +**

Label both ends before you strip.

02 XLR-3 balanced audio
03 XLR connector faces
04 Euroblock & TRS / TS
05 Speakers: speakON & 70 V
06 DMX pinouts (3 & 5-pin)
07 Belden 9729: which wire
08 9729 termination kit
09 DMX over Cat (E1.27-2)
10 Ethernet T568A / B
11 LC fiber & strand colors
12 Coax: RG6 / RG8 / RG8X
13 Control: RS-232 / 485
14 Closeout checklist
15 DMX termination kit (illustrated)

Device-specific terminations & updates

illuminated-integration.github.io/field-terminations

illuminated-integration.com
717 996 4596 · Harrisburg PA

XLR-3 BALANCED AUDIO

AUDIO
02 / 15

XLR-3 PINOUT (AES14)

PIN	SIGNAL	CONDUCTOR
1	**Shield / Gnd**	drain / braid
2	**Hot (+)**	colored / light
3	**Cold (–)**	black / dark

1-SHIELD · 2-HOT · 3-COLD

- Read the pin numbers molded beside each solder cup; don't rely on face orientation from memory.
- Sleeve the drain before it reaches pin 1 so it can't touch pin 2.

PAIR COLORS · HOUSE STANDARD

PAIR	PIN 2 (+)	PIN 3 (–)
Red / Black	Red	Black
Red / White	Red	White
White / Black	White	Black

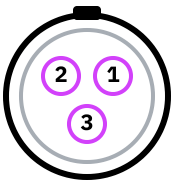
- Shield to pin 1 at both ends. Leave the shell lug open unless the spec says otherwise.
- Match what's already on the job before mixing standards.
- Never use mic cable for DMX (see card 06).

XLR CONNECTOR FACES

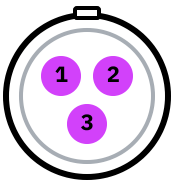
AUDIO · DMX

03 / 15

XLR-3 · FRONT VIEW, LATCH UP



FEMALE
socket face



MALE
pin face

PIN	AUDIO	DMX
1	Shield	Common
2	**Hot +**	**Data –**
3	**Cold –**	**Data +**

Same connector, opposite polarity.
Pin 2 is **+** for audio and **–** for DMX.
Never re-use an audio cable for DMX.

- Solder side is the mirror image of the front view.
- The numbers molded on the insert are the final word.
- XLR-5 pinout: card 06.

EUROBLOCK & TRS / TS

AUDIO

04 / 15

EUROBLOCK / PHOENIX, 3-POSITION



XLR pin 2 · XLR pin 3 · XLR pin 1

- **Order varies by manufacturer.** Follow the silkscreen, not habit.
- Strip to the device's spec (typ. 5–7 mm). No bare copper past the clamp.
- **Don't tin wire** for screw terminals; solder creeps and loosens. Ferrules if they fit.
- Tug-test every wire. Heat-shrink any lifted shield.

¼" / 3.5 MM TRS · BALANCED OR STEREO



SLEEVE Shield · **RING** Cold – · R · **TIP** Hot + · L

TS · UNBALANCED



SLEEVE Ground · **TIP** Signal

Balanced out → unbalanced in: + to tip; – and shield to sleeve at the unbalanced end. Some outputs must not have – grounded; check the manual.

SPEAKERS: SPEAKON & 70 V

AUDIO
05 / 15

SPEAKON NL4 / NL2

CONTACT	FULL-RANGE	BI-AMP*
1+	+	LF +
1–	–	LF –
2+	—	HF +
2–	—	HF –

*Common, not universal. NL2 = 1± only. Check the speaker manual.

SPEAKER WIRE POLARITY

- Red / Black jacketed: **Red = +**, Black = –. **NEVER** use Ferrules!

70 V / 100 V LINES

- Line – to transformer COM; line + to the wattage tap.
- Verify tap setting from drawing specification.
- Never land a 70 V line on the 8 Ω tap. Don't mix 70 V and 100 V taps.
- Polarity check: tester, or a 9 V pop on a passive woofer (+ to +, cone moves out).

DMX PINOUTS: 3 & 5-PIN

LIGHTING
06 / 15

DMX512 XLR (E1.11)

PIN	SIGNAL
1	**Common / Shield**
2	**Data 1 –**
3	**Data 1 +**
4	Data 2 – (spare)
5	Data 2 + (spare)

1-SHIELD · 2-DATA– · 3-DATA+

- **Daisy-chain only.** No Y-cables; use an opto-splitter.
- **Terminate** the last device: 120 Ω across pins 2 & 3.
- 32 unit loads per segment, max.
- DMX needs 120 Ω cable. **Mic cable will cause errors.**
- 3-pin is common but non-standard. 3↔5 adapters run pin-for-pin 1-2-3.

BELDEN 9729: WHICH WIRE?

PAIR 1 · Black & Red
Always the primary DMX
- Red → Data +
- Black → Data –
- Drain → Common

PAIR 2 · Black & White
5-pin spare, else fold back
- White → Data 2 +
- Black → Data 2 –
- Drain → Common

WIRE	SIGNAL	XLR
Red ①	**Data +**	3
Black ①	**Data –**	2
Drain ①	**Common**	1
White ②	Data 2 +	5
Black ②	Data 2 –	4
Drain ②	**Common**	1

TWO BLACK WIRES. Keep each black with its own foil and twist. The black twisted with **red** is Data –. Pair 2 black only ever goes to pin 4.

Block = push-in IN/THRU terminal (card 08). Pair 2 on pins 4/5 is house standard.

9729 → PUSH-IN TERMINAL KIT

Kit: HS1 ×2 drain sleeve · HS2 ×1 jacket · HS3 ×2 foil · HS4 ×1 spare-pair cap · 2 cable ties. **Tools:** stripper, flush cutters, heat gun, 3 mm flat blade.

1. Label the cable (DMX #). Strip **190 mm** of jacket.
2. Separate the pairs. Cut foils back to **10 mm** from the jacket.
3. **HS1** over both drains, leaving 10 mm bare. Shrink.
4. **HS3** over each foil stub. Shrink.
5. **HS2** over the jacket end and stubs. Shrink.
6. Fold the spare pair back, cap with **HS4**, tie it to the jacket. Tie the working pair 30 mm from its ends.
7. Strip **10 mm**: Red → 3 / 6, Black → 2 / 5, Drain → 1 / 4.
8. Push a **3 mm** blade into the release (A), insert the wire (B), pull the blade, tug-test.
9. IN only: 1-2-3. IN/THRU: incoming on 1-2-3, outgoing on 4-5-6.

Illustrated steps for 9729 and Cat cable on a screw-down block: **card 15**.

DMX OVER CAT CABLE

LIGHTING
09 / 15

RJ45 ↔ XLR · ANSI E1.27-2, T568B

RJ45	WIRE	SIGNAL	XLR
1	Wh / Org	**Data 1 +**	3
2	Orange	**Data 1 –**	2
3	Wh / Grn	Data 2 +	5
6	Green	Data 2 –	4
7	Wh / Brn	**Common**	1
8	Brown	**Common**	1
4, 5	Blue pair	not used	—

- Cat5e or better, UTP or STP.
- **Not every vendor follows E1.27-2.** Some pixel / show-control gear and some IDC kits use their own RJ45 map. Check the manual.
- DMX on Cat cable is **not Ethernet**. Label it DMX and never patch it into a switch.

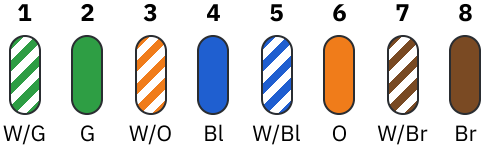
ETHERNET: T568A / T568B

NETWORK
10 / 15

T568B · house default



T568A



Plug view: contacts toward you, latch away, **pin 1 on the left**.

- Same standard both ends = straight-through.
- Untwist **≤ 13 mm (½")**. Jacket goes under the crimp strain relief.
- Pass-through plugs: trim flush, no copper whiskers.
- Shielded: bond the drain to the plug or jack shield.
- 100 m channel max (90 m permanent + 10 m patch).
- Wiremap-test every run. Certify all runs carrying Dante, HDBaseT, or PoE.

LC FIBER & STRAND COLORS

FIBER
11 / 15

POLARITY



A-to-B duplex: TX always lands on RX.

- No link? Flip the duplex clip / uniboot polarity.
- **Inspect → clean → inspect** before every mate.
- **Never look into a port or fiber end.** Use a VFL.
- No kinks or crushed ties; use hook-and-loop.

CONNECTOR BODY COLOR (TIA)

- Singlemode UPC
- **Singlemode APC** (8°)
- OM1 62.5 µm
- OM2 50 µm
- OM3 / OM4 (OM4 may be violet)
- OM5

Never mate APC to UPC. Green only goes to green.

STRAND & TUBE ORDER · TIA-598

13–24 repeat with a black tracer

1	2	3	4	5	6	7	8	9	10	11	12
Blue	Orange	Green	Brown	Slate	White	Red	Black	Yellow	Violet	Rose	Aqua

COAX: RG6, RG8 & RG8X

VIDEO · RF
12 / 15

AT A GLANCE

	RG6	RG8	RG8X
Impedance	**75 Ω**	**50 Ω**	**50 Ω**
Size	~0.27" OD	~0.40" OD	~0.24" OD
Connectors	F, BNC 75	N, BNC 50, UHF	BNC 50, UHF

Never mix 50 Ω and 75 Ω. BNC comes in both: 75 Ω for SDI video, 50 Ω for wireless antennas. Check the marking.

- Use the compression connector made for that exact cable (RG6 and RG6 quad differ).
- No braid strands near the center pin.

RG6 75 Ω

- **SDI video** (HD / 3G / 12G) on precision RG6
- CATV, off-air antenna, satellite

RG8 50 Ω

- **Wireless mic & IEM antenna runs** (remote paddles)
- Cellular / Wi-Fi antenna feeds (LMR-400 class)

RG8X 50 Ω

- Short antenna jumpers where RG8 is too stiff to dress
- Higher loss than RG8: keep runs short

Long antenna runs: lower loss beats thinner cable. **VERIFY** from manufacturer impedance required of cable.

CONTROL: RS-232 / 485 / GPIO

RS-232 · DB9 (DTE / COMPUTER SIDE)

DB9	SIGNAL	LANDS ON
2	RX	far-end TX
3	TX	far-end RX
5	Ground	Ground

- Phoenix Tx / Rx / G: **cross Tx↔Rx**, ground to ground.
- Most AV control uses 3 wires. No handshake unless the protocol doc asks for it.
- No response? Swap Tx/Rx first, then check baud / parity / stop bits.

RS-485 / 422

- Data on one twisted pair, plus common. **A/B naming isn't consistent** between vendors; if it's dead, swap A/B.

RELAYS & GPIO

- Relay: COM + **NO** (open at rest) or **NC** (closed at rest).
- Confirm GPIO voltage and pull-up before landing a contact closure.

CLOSEOUT CHECKLIST

BEFORE YOU LEAVE THE RACK

- [ ] Both ends labeled to the drawing's cable ID
- [ ] Patch panels labeled per drawing specifications
- [ ] Continuity + polarity tested (audio / speaker)
- [ ] DMX: last device terminated
- [ ] Cat: wiremap pass, certified where required
- [ ] Fiber: inspected, cleaned, light-level checked
- [ ] Coax: correct impedance connectors, signal verified
- [ ] Strain relief, service loop, no copper showing
- [ ] as-builts updated

Device-specific pinouts can be found here. Missing something? Submit a request!

JOB NOTES

Belden 9729

DMX Cable Termination Kit With Screw Down Connector

Category Cable

